A piecewise-linear function as a shallow ReLU network

One hidden layer, one hinge per neuron, placed by one number

A continuous piecewise-linear function with K kinks is a line plus K ReLU hinges. The note *A piecewise-linear function as a deep, narrow ReLU network* builds such a function one kink per layer, with two neurons per layer: a *hinge-chain network*. The *hinge-ladder network* uses three neurons per layer, so that every hinge reads x itself. This note takes the opposite road: a single hidden layer, as wide as needed. Each hidden neuron is a hinge $\mathrm{ReLU}(x - t_j)$, set by one number, its position t_j, and the output neuron adds the hinges up with weights w_j and a bias c. The input fans out to the hinges and the hinges fan back in to the output, so we call it a *hinge-fan network*.

We show what it computes (Section 1) and that the hinge form of a piecewise-linear function is such a network, with one hinge per kink and one more for the first slope (Section 2). Fixing every input weight at 1 costs at most one neuron against general ReLU neurons (Section 3). With the hinges fixed, the output neuron does linear regression on the hinges, so its weights solve a least-squares problem (Section 4). Section 5 counts the hinges that a smooth function needs, on the fifteen targets of the course's pages, and Section 6 sets the three networks side by side. You need to know what a ReLU neuron is and what least squares does.

§1 The network

Throughout, $\mathrm{ReLU}(z) = \max\{z, 0\}$ and x ranges over an interval $[a, b]$; every example uses $[a, b] = [-2, 2]$, the domain of the pages. A hinge-fan network with n hinges has one input x, one hidden layer of n ReLU neurons, the *hinges*, and one linear output y:

$$h_j(x) = \mathrm{ReLU}(x - t_j) \qquad (j = 1, \dots, n), \tag{1}$$

$$y(x) = c + \sum_{j=1}^{n} w_j \, h_j(x). \tag{2}$$

Hinge j has the input weight 1, fixed, and the bias $-t_j$. So its one knob is its *position* $t_j \in [a, b]$, where it bends: it is 0 left of t_j and rises with slope 1 right of it. Every hinge opens to the right. The output neuron has two kinds of knobs: the *weight* w_j on the arrow from hinge j, and the *bias* c. That makes $2n + 1$ free numbers. Figure 1 draws the network.

Proposition 1 (What the network computes). *On $[a, b]$, the output y of (2) is continuous and piecewise linear, and:*

(a) it can bend only at the positions t_j inside (a, b), and its slope at any other point x is $\sum_{j:\, t_j < x} w_j$, so that crossing a position the slope jumps by the weights of the hinges there;

(b) $y = c$ on $[a, \min_j t_j]$: left of every hinge the curve is flat at height c, and in particular $y(a) = c$;

(c) a hinge at $t_j = a$ is the line $x - a$ on the whole of $[a, b]$, and a hinge at $t_j = b$ is 0 there.

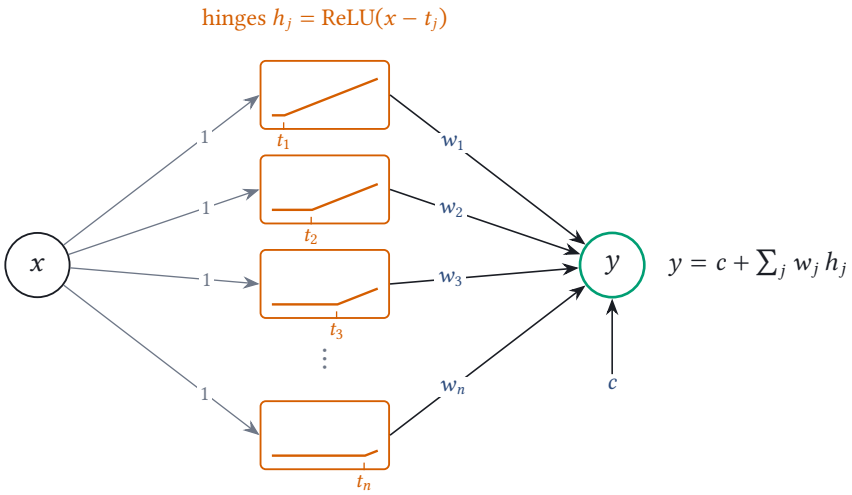


Figure 1: A hinge-fan network with n hinges. Every arrow out of x has the fixed weight 1, so hinge j only adds its bias $-t_j$ and applies the ReLU; its box draws its output for x from a to b, flat up to t_j and rising with slope 1 after it. The free numbers are the positions t_j, the output weights w_j and the bias c.

Proof. Each h_j is continuous and piecewise linear, with slope 0 left of t_j and 1 right of it, so y is continuous and piecewise linear and can bend only where some h_j does. At a point x that is no position, $h_j'(x)$ is 1 if $t_j < x$ and 0 if $t_j > x$, which gives the slope in (a). If $x \le t_j$ for every j, all hinges vanish and $y(x) = c$, which is (b). For (c), $x - a \ge 0$ and $x - b \le 0$ on $[a, b]$. □

So every knob has a plain meaning: w_j turns the curve at t_j, c is the height at which the curve starts, and a hinge parked at a tilts the start.

§2 The hinge form is the network

Let $f : [a, b] \to \mathbb{R}$ be continuous and piecewise linear, with $K \ge 0$ kinks, the points where its slope changes:

$$a < t_1 < t_2 < \cdots < t_K < b.$$

Put $t_0 = a$ and $t_{K+1} = b$, let m_i be the slope of f on $[t_i, t_{i+1}]$ for $i = 0, \dots, K$, and let $\Delta_i = m_i - m_{i-1}$ be the change of slope at the ith kink. (The narrow note writes s_i for the slopes.)

Proposition 2 (The hinge form is a hinge-fan network). *For every $x \in [a, b]$,*

$$f(x) = f(a) + m_0\ \mathrm{ReLU}(x - a) + \sum_{i=1}^{K} \Delta_i\ \mathrm{ReLU}(x - t_i). \tag{3}$$

So f is a hinge-fan network with $K + 1$ hinges, numbered 0 to K: hinge 0 at $t_0 = a$ with $w_0 = m_0$, hinge i at the kink t_i with $w_i = \Delta_i$, and the bias $c = f(a)$. If f starts flat ($m_0 = 0$), hinge 0 has weight 0 and can go: then K hinges suffice.

Proof. Both sides are continuous and piecewise linear on $[a, b]$. At $x = a$ every hinge vanishes, so the right side is $f(a)$. Between t_i and t_{i+1} its slope is $m_0 + \Delta_1 + \cdots + \Delta_i = m_i$ by Proposition 1(a), because the sum telescopes; that is the slope of f there. Two continuous functions that agree at a and have the same slope on every piece agree on all of $[a, b]$. □

Equation (3) is the hinge form of the narrow note, whose hinge 0 at $t_0 = a$ carries $\Delta_0 = m_0$. Read as a network it needs nothing more, and every number of the network is a number of f: where it bends, by how much, and where it starts. The count cannot be beaten.

Proposition 3 (What n hinges can make). *On $[a, b]$, the hinge-fan networks with n hinges make exactly the continuous piecewise-linear functions with at most $n - 1$ kinks, together with those with n kinks whose first piece is flat.*

Proof. A function of either kind is a network with at most n hinges by Proposition 2, and any spare hinge can sit anywhere with weight 0. Conversely, a network with n hinges bends only at its positions inside (a, b) (Proposition 1), so it has at most n kinks. If it has n, its n positions are distinct and lie inside (a, b), so none sits at a, and the first piece is flat by Proposition 1(b). □

So K kinks take $K + 1$ hinges, or K when f starts flat, and no network with fewer hinges makes f, wherever its hinges sit.

Example: the zigzag. The zigzag of the pages passes through $(-2, -1)$, $(-1, 1)$, $(0, 0)$, $(1, 2)$ and $(2, -1)$. It has $K = 3$ kinks, at -1, 0 and 1, the slopes $m = (2, -1, 2, -3)$, the slope changes $(\Delta_1, \Delta_2, \Delta_3) = (-3, 3, -5)$, and $f(a) = -1$. Proposition 2 gives the network with four hinges

$$y = -1 + 2\,\mathrm{ReLU}(x + 2) - 3\,\mathrm{ReLU}(x + 1) + 3\,\mathrm{ReLU}(x) - 5\,\mathrm{ReLU}(x - 1),$$

that is $c = -1$, $t = (-2, -1, 0, 1)$ and $w = (2, -3, 3, -5)$. The bias and hinge 0 together make the first piece, $2x + 3$. Table 1 lists every term at the points $x = -2, \dots, 2$, and Figure 2 draws the terms and their running sums. The staircase of the pages starts flat, at height $-3/2$, and has six kinks, so six hinges make it, none of them at a: $c = -3/2$, $t = (-1.3, -1.1, -0.2, 0, 0.9, 1.1)$ and $w = (5, -5, 5, -5, 5, -5)$.

x	-2	-1	0	1	2
c	-1	-1	-1	-1	-1
$2\,\mathrm{ReLU}(x + 2)$	0	2	4	6	8
$-3\,\mathrm{ReLU}(x + 1)$	0	0	-3	-6	-9
$3\,\mathrm{ReLU}(x)$	0	0	0	3	6
$-5\,\mathrm{ReLU}(x - 1)$	0	0	0	0	-5
y	-1	1	0	2	-1
f	-1	1	0	2	-1

Table 1: The zigzag's network at the points $x = -2, \dots, 2$: the bias, the four weighted hinges, and their sum y, which equals the target f.

§3 Why the position is the only knob a hinge needs

A general ReLU neuron with one input is $\mathrm{ReLU}(ux + v)$, with an input weight u and a bias v. Two facts about the ReLU turn it into a hinge:

$$\mathrm{ReLU}(\lambda z) = \lambda\ \mathrm{ReLU}(z) \quad (\lambda \geq 0), \qquad \mathrm{ReLU}(z) - \mathrm{ReLU}(-z) = z. \tag{4}$$

The first moves a positive factor out of the ReLU and onto the next weight. The second says that a hinge that opens to the left is a hinge that opens to the right minus a line.

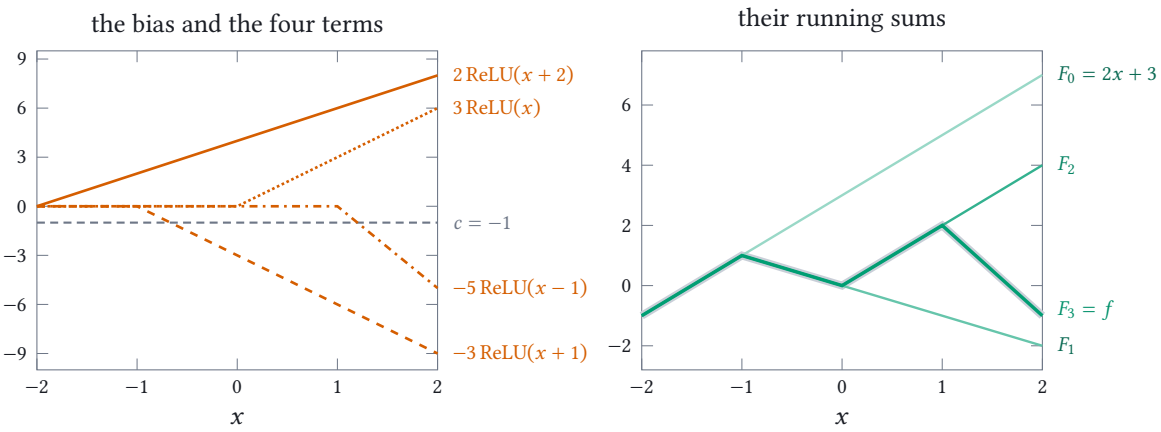


Figure 2: The zigzag as a sum of hinges. *Left:* the bias $c = -1$ and the four weighted hinges; each is flat up to its position and straight after it. *Right:* the running sums $F_\ell = c + \sum_{i \le \ell} w_i \,\mathrm{ReLU}(x - t_i)$, over the zigzag f (thick grey). F_0 is the first piece, and each further hinge turns the curve at its kink; $F_3 = f$.

Proposition 4 (One number per hinge costs at most one neuron). *Let $g(x) = d + \sum_{j=1}^{n} r_j \,\mathrm{ReLU}(u_j x + v_j)$ be a network with one hidden layer of n arbitrary ReLU neurons. On $[a, b]$, g is a hinge-fan network with at most $n + 1$ hinges: one at a, and at most one for each neuron of g.*

Proof. Take one term $r \,\mathrm{ReLU}(ux + v)$ and, when $u \neq 0$, put $\tau = -v/u$, the point where the neuron bends.

- If $u > 0$, then $\mathrm{ReLU}(ux + v) = \mathrm{ReLU}\big(u(x - \tau)\big) = u \,\mathrm{ReLU}(x - \tau)$ by (4): the term is a hinge at τ with weight ru.
- If $u < 0$, then $\mathrm{ReLU}(ux + v) = |u| \,\mathrm{ReLU}(\tau - x)$, and (4) with $z = x - \tau$ gives $\mathrm{ReLU}(\tau - x) = \mathrm{ReLU}(x - \tau) - (x - \tau)$. On $[a, b]$ the line is $x - \tau = \mathrm{ReLU}(x - a) + (a - \tau)$. So the term is a hinge at τ with weight $r|u|$, minus $r|u|$ times the hinge at a, plus a constant.
- If $u = 0$, the term is the constant $r \,\mathrm{ReLU}(v)$.

A hinge at a position outside $[a, b]$ brings nothing new: if $\tau \ge b$ it is 0 on $[a, b]$, and if $\tau \le a$ it equals $\mathrm{ReLU}(x - a) + (a - \tau)$ there, the hinge at a plus a constant. Adding up, each neuron of g contributes at most one hinge inside (a, b), a multiple of the hinge at a, and a constant. The constants go into c and the multiples into the weight of the hinge at a. □

The extra hinge is sometimes needed. On $[-2, 2]$ the single neuron $\mathrm{ReLU}(-x)$ has one kink, at 0, and the first slope -1, so by Proposition 3 it takes two hinges: $\mathrm{ReLU}(-x) = 2 - \mathrm{ReLU}(x + 2) + \mathrm{ReLU}(x)$. (On the 801 points of the pages, the best single hinge leaves a mean squared error of 0.084.) Two consequences. A position never needs to leave $[a, b]$, so a slider for a position can stay there. And fixing the input weights at 1 removes a redundancy of general neurons: (u, v, r) and $(\lambda u, \lambda v, r/\lambda)$ give the same term for every $\lambda > 0$.

§4 With the hinges fixed, the slopes are least squares

To fit a network to a target f, we compare the two at points $x_1, \ldots, x_N$ of $[a, b]$; the pages and the tables below use the $N = 801$ equally spaced points of $[-2, 2]$ and the mean squared error

$$E = \frac{1}{N} \sum_{i=1}^{N} \big(y(x_i) - f(x_i)\big)^2.$$

Fix the positions $t_1, \dots, t_n$. The hinges are then fixed functions, the *features* $\varphi_j(x) = \mathrm{ReLU}(x - t_j)$, and $y = c + \sum_j w_j \varphi_j$ is linear in $\theta = (c, w_1, \dots, w_n)$. Let Φ be the $N \times (n+1)$ matrix whose ith row is $\big(1, \varphi_1(x_i), \dots, \varphi_n(x_i)\big)$, and $\mathbf{f} = \big(f(x_1), \dots, f(x_N)\big)$. Then $E(\theta) = \frac{1}{N}\|\Phi\theta - \mathbf{f}\|^2$ is a convex quadratic in θ, and its minimizers are the solutions of the normal equations

$$\Phi^\top \Phi\, \theta = \Phi^\top \mathbf{f}. \tag{5}$$

This is linear regression, as in the regression part of the course, with the hinges in place of the inputs: the hidden layer chooses the features and the output neuron regresses on them. The note *Kernelized polynomial regression* does the same with monomials and then with Gaussian features. The note *Random kitchen sinks* draws the hidden units at random and fits only the output layer; with ReLU units, Proposition 4 makes such a model a hinge-fan network whose hinges sit at random positions, plus one at a.

Lemma 1. *Let $a \le t_1 < \cdots < t_n < b$. If some x_i lies in $[a, t_1]$ and at least one lies strictly between any two consecutive positions and in $(t_n, b]$, then Φ has full column rank, and* (5) *has exactly one solution.*

Proof. Let $\Phi\theta = 0$, so that $p = c + \sum_j w_j \varphi_j$ vanishes at every x_i. On $[a, t_1]$, $p = c$, so $c = 0$. Suppose that $p = 0$ on $[a, t_j]$ and $w_1 = \cdots = w_{j-1} = 0$. On $[t_j, t_{j+1}]$ (with $t_{n+1} = b$) the function p is linear with slope w_j and $p(t_j) = 0$, and it vanishes at a point strictly to the right of t_j, so $w_j = 0$ and $p = 0$ up to t_{j+1}. By induction $\theta = 0$. □

On the 801 points, distinct positions on the grid of spacing 0.01 below b, which is where the fit below puts them, always satisfy the lemma.

Moving the hinges is the hard part. The error is convex in θ only. Write $E^*(t) = \min_\theta E(t, \theta)$ for the best error with the hinges at $t = (t_1, \dots, t_n)$: the landscape that moving the hinges explores. For the target $|x| - 1$ of the pages and $n = 2$, Proposition 2 gives $E^*(-2, 0) = E^*(0, -2) = 0$, since the two hinges merely swap. At the midpoint $(-1, -1)$ both hinges sit at -1, the curve can bend only there, and $E^*(-1, -1) = 0.33$. A nonnegative convex function vanishes at the midpoint of two zeros, so E^* is nonconvex, and so is E jointly in (t, θ): minimizing a jointly convex function over θ would leave a convex function of t.

The fit of this note. The tables and figures below place n hinges in two ways and keep the better. The kinks come from the dynamic program of the pages: among the 401 points of $[-2, 2]$ with spacing 0.01, choose the k interior kinks for which the piecewise-linear interpolant of f through a, the kinks and b has the least squared error at those points; dynamic programming over the last kink finds them exactly. Placement *left* puts a hinge at a and the other $n - 1$ at the best $n - 1$ kinks. Placement *flat* puts all n at the best n kinks and none at a, which pays off when f starts flat. For each, c and w solve (5), and the placement with the smaller error wins. The kinks of the menu's targets lie on the grid, so this fit is exact on every piecewise-linear target as soon as Proposition 3 allows it. Gradient steps on the positions can still lower the error: fifty of them take the Runge function's error with seven hinges from 1.3×10^{-4} to 1.1×10^{-4}.

Placed against evenly spread. Hinges spread evenly over $[a, b]$, the first at a, and never moved, form a fixed basis, the linear splines on a uniform grid, and only the slopes are fitted. They need a few more hinges on a smooth target, and many more on a local feature, which needs a hinge where it is (Figure 3). Five evenly spread hinges miss the peak of the Runge function $1/(1 + 8x^2)$ and leave a maximum error of 0.40. Five placed ones sit at 0, ± 0.53 and ± 0.97, none at a, and leave 0.052. Evenly spread hinges make a piecewise-linear target exactly only once their grid hits every kink: the steep step, with kinks at ± 0.2, takes 20 of them against 2 placed ones, and the staircase 40 against 6 (Table 2). Placing the hinges is what the hidden layer learns.

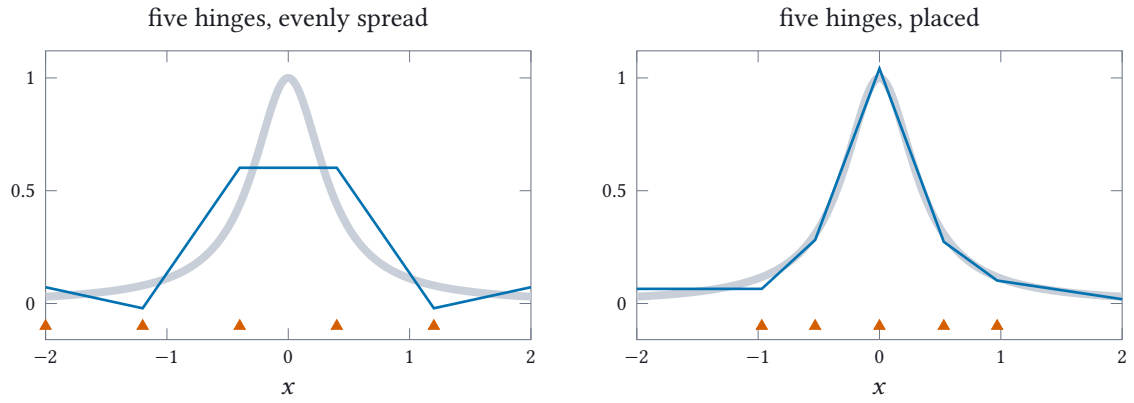


Figure 3: The Runge function $1/(1+8x^2)$ (thick grey) fitted with five hinges, whose positions the red triangles mark, with the slopes and the bias by least squares. *Left:* evenly spread from -2, they miss the peak: mean squared error 1.9×10^{-2}, maximum error 0.40. *Right:* placed by the fit of Section 4, all five near the peak and none at a: mean squared error 3.6×10^{-4}, maximum error 0.052.

§5 How many hinges a smooth function needs

Proposition 5 (Linear interpolation). *Let f be twice differentiable on $[a,b]$ with $|f''|\le M$, and let p be the piecewise-linear interpolant of f at the $n+1$ equally spaced points $x_k=a+kh$, $h=(b-a)/n$. Then $|f(x)-p(x)|\le Mh^2/8$ for every $x\in[a,b]$.*

Proof. Fix x strictly between two nodes x_k and x_{k+1}, put $e=f-p$, which vanishes at both nodes, and $q(s)=(s-x_k)(s-x_{k+1})$, and choose λ with $e(x)=\lambda q(x)$. The function $e-\lambda q$ vanishes at x_k, x and x_{k+1}, so by Rolle's theorem its derivative vanishes at two points and its second derivative at some ξ between them. Since $p''=0$ between the nodes and $q''=2$, this says $f''(\xi)=2\lambda$, and so $|e(x)|=\frac12|f''(\xi)|\,|q(x)|\le\frac12M\cdot\frac14h^2$. □

The interpolant has n pieces, hence $n-1$ interior kinks, so by Proposition 2 it is a hinge-fan network with n hinges, the first at a. So n hinges always get within

$$\frac{M(b-a)^2}{8n^2}\qquad\left(=\frac{2M}{n^2}\ \text{ on }[-2,2]\right)\tag{6}$$

of f everywhere, and least squares with the same hinges has a mean squared error at most the square of (6), on any points. The error falls like n^{-2} and the mean squared error like n^{-4}.

Tables 2 and 3 run the fit of Section 4 on the fifteen targets of the pages. Each piecewise-linear target is exact with $K+1$ hinges, or K when it starts flat, and never with fewer (Proposition 3). For each smooth target we count the hinges until the maximum error on the 801 points drops below 0.02, about one percent of the height of most targets: by the bound (6), that is the least n with $2M/n^2<0.02$; with evenly spread hinges; and with placed ones. Placed hinges always beat the bound, by far when the curvature is local: the Runge function, whose $|f''|$ reaches 16 at $x=0$ and stays below 0.51 for $|x|\ge1$, takes 41 hinges by the bound, 29 evenly spread and 9 placed. Figure 4 shows the chirp with 6, 12 and 24 hinges, and the mean squared error against n: beyond a few hinges, every smooth target follows the slope -4 of the bound on log-log axes. The sine, for instance, goes from 1.7×10^{-4} at $n=8$ to 1.0×10^{-5} at $n=16$.

Least squares against interpolation. The hinge-ladder page fits with the same kinks, but interpolates f at them. Where f is convex the interpolant lies above f on the whole piece, and least squares lowers it until it crosses f. With ten hinges this divides the mean squared error by about 4 to 6 on the seven

			hinges for an exact fit	
Target	kinks K	first slope m_0	placed	evenly spread
Zigzag	3	2	4	4
Absolute value	1	-1	2	2
Tent	3	0	3	4
W shape	3	-2	4	4
Clipped ramp	2	0	2	8
Steep step	2	0	2	20
Triangle wave	7	3	8	8
Staircase	6	0	6	40

Table 2: The eight piecewise-linear targets of the pages. Placed hinges make each one exactly with $K + 1$ hinges, or K when $m_0 = 0$, as Proposition 3 demands. Evenly spread hinges, the first at a, make it exactly once their grid contains every kink.

			hinges for a maximum error below 0.02			MSE below 10^{-4}
Target	$f(x)$	$M = \max \lvert f'' \rvert$	bound	evenly spread	placed	placed
Sine	$\sin(\pi x/2)$	2.47	16	13	11	9
Two periods	$\sin(\pi x)$	9.87	32	26	21	18
Parabola	$x^2/2 - 1$	1.00	11	9	9	8
Cubic	$x^3/4 - x/2$	3.00	18	14	11	9
Gaussian bump	$\frac{3}{2}\, e^{-3x^2/2}$	4.50	22	17	11	9
Runge function	$1/(1 + 8x^2)$	16.00	41	29	9	8
Chirp	$\sin(\frac{3}{2}x^2)$	29.62	55	44	24	19

Table 3: The seven smooth targets of the pages, on $[-2, 2]$. *Bound*: the least n with $2M/n^2 < 0.02$, from (6). *Evenly spread* and *placed*: the least n for which the least-squares fit with evenly spread hinges, or the fit of Section 4, has a maximum error below 0.02 on the 801 points. The last column asks for a mean squared error below 10^{-4}.

smooth targets: 7.0×10^{-5} against 3.6×10^{-4} for the sine (a factor 5.1), 8.7×10^{-4} against 3.4×10^{-3} for two periods (3.9), and 1.0×10^{-3} against 4.2×10^{-3} for the chirp (4.0); the factors of the parabola, the cubic, the bump and the Runge function are 6.0, 5.1, 4.3 and 4.4. The two networks make the same functions (Section 6), so the whole difference lies in the fit.

§6 Three networks for one function

Table 4 sets the hinge-fan network beside the hinge chain of the narrow note and the hinge ladder. All three make a piecewise-linear function with K kinks exactly, and their counts match: each kink costs one layer of a deep network or one hinge of the fan. The ladder and the fan spend one more on the first slope, which the fan saves when f starts flat.

The ladder, folded flat. The hinge ladder computes

$$y = x + b_0 + \sum_{\ell} w_\ell \, \mathrm{ReLU}(x + b_\ell),$$

a running sum that gains one hinge per layer, at $-b_\ell$. Since $x = \mathrm{ReLU}(x - a) + a$ on $[a, b]$, this is a hinge-fan network with the hinges $t_\ell = -b_\ell$ and the weights w_ℓ, one more unit of weight on the hinge

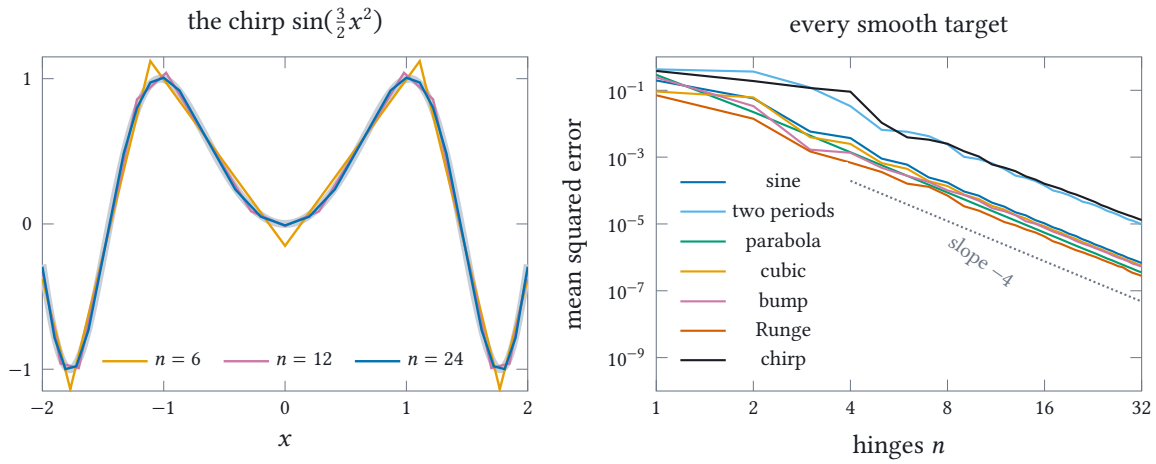


Figure 4: *Left:* the chirp (thick grey) fitted by the fit of Section 4 with 6, 12 and 24 hinges; the mean squared error drops from 3.9×10^{-3} to 4.9×10^{-4} and 3.7×10^{-5}, and the maximum error from 0.16 to 0.069 and below 0.02. *Right:* the mean squared error of the fit against the number of hinges, for the seven smooth targets; the dotted line has slope −4, the rate of (6).

	Hinge chain	**Hinge ladder**	**Hinge fan**
Hidden layers	K	$K + 1$	1
Neurons per hidden layer	2: a hinge and a running sum	3: a sum, a hinge and a copy of x	$K + 1$ hinges, or K if f starts flat
A hinge reads	the hinge before it	the copy of x	x
ReLU on	every neuron, with a lift C	the hinges only	the hinges only
Free numbers	5 per layer, 3 at the output	w_ℓ, b_ℓ per layer, and b_0	t_j, w_j per hinge, and c
The kinks	in order, by their gaps	anywhere, in any order	anywhere, in any order

Table 4: Three networks that make a continuous piecewise-linear function with K kinks exactly.

at a (merged with a layer that sits there), and $c = b_0 + a$. Conversely, a hinge-fan network is a ladder with one layer per hinge, the layer at a carrying $w_0 - 1$; a layer at a with weight -1 is added when no hinge sits there. The ladder's running sum collects the hinges one layer at a time, and the fan's output collects them all at once.

What depth buys. For these functions, nothing: a kink costs one layer of a deep network or one hinge of the shallow one. Depth pays when layers compose. The tent map $T(u) = 2\,\mathrm{ReLU}(u) - 4\,\mathrm{ReLU}(u - \frac{1}{2})$ maps $[0, 1]$ onto itself, and its L-fold composition is a sawtooth with 2^L pieces of slopes $\pm 2^L$. A network with L hidden layers of two ReLU neurons computes it, while a hinge-fan network needs 2^L hinges: the sawtooth has $2^L - 1$ kinks and starts with the slope 2^L (Proposition 3).[1]

The page `docs/visualizations/hinge-fan-networks/` draws these networks on the fifteen targets: drag the hinges, turn the slopes, and compare with its fit, which tries the two placements of Section 4 and also the mirror image of the flat one.

[1] M. Telgarsky, *Benefits of depth in neural networks*, Conference on Learning Theory (COLT), 2016. He turns this sawtooth into a lower bound: networks with far fewer layers need exponentially many neurons even to approximate it.